

Experiment: Bank Rate of Interest Using Method Overriding

Experiment No: 02

Date: 17/09/2026

Title: Bank Rate of Interest using Inheritance and Method Overriding

Objective / Aim

To write and execute a Java program to find the **rate of interest of different banks** using inheritance, method overriding, and runtime polymorphism. The program demonstrates how different bank classes such as SBI, ICICI, and PNB override the rate() method of the parent class RBI.

System Requirements

Hardware

- Standard PC/Laptop
- Minimum 4 GB RAM
- Keyboard and Monitor

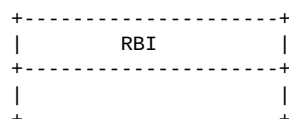
Software

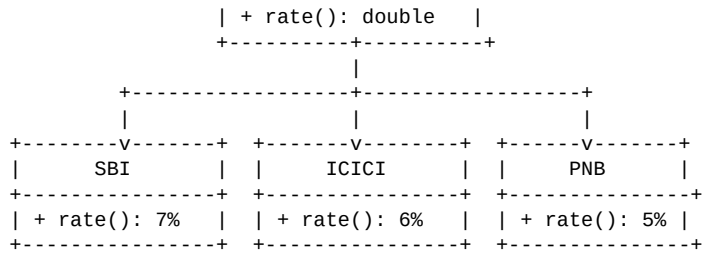
- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK 8 or above)
- Java-compatible IDE or text editor
- Command Prompt / Terminal

Algorithm / Logic Description

1. Start the program.
2. Import the java.util.* package to use the Scanner class.
3. Create a parent class named RBI.
4. Define a rate() method in RBI that returns an interest rate of 4%.
5. Create SBI, ICICI, and PNB classes by extending RBI.
6. Override rate() in SBI to return 7%, ICICI to return 6%, and PNB to return 5%.
7. Create a Scanner object to accept the bank name from the user.
8. Declare an RBI reference variable named b.
9. Compare the entered bank name using equalsIgnoreCase().
10. Create the corresponding bank object. Any input other than RBI, SBI, or ICICI selects PNB.
11. Call rate() using the RBI reference. Method overriding selects the appropriate implementation.
12. Display the rate of interest and stop the program.

UML Class Diagram





Inheritance Relationship

```

SBI ----- inherits from -----> RBI
ICICI ----- inherits from -----> RBI
PNB ----- inherits from -----> RBI

```

Method Overriding

Bank	Rate of Interest
RBI	4%
SBI	7%
ICICI	6%
PNB	5%

Source Code

```

import java.util.*;

class RBI {
    double rate() {
        return 4;
    }
}

class SBI extends RBI {
    double rate() {
        return 7;
    }
}

class ICICI extends RBI {
    double rate() {
        return 6;
    }
}

class PNB extends RBI {
    double rate() {
        return 5;
    }
}

public class Bank {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        System.out.printf(
            "Enter the Bank name to find the rate of Interest : "
        );

        String s = sc.next();

        RBI b;

        if(s.equalsIgnoreCase("RBI"))
            b = new RBI();
        else if(s.equalsIgnoreCase("SBI"))
            b = new SBI();
    }
}

```

```

        else if(s.equalsIgnoreCase("ICICI"))
            b = new ICICI();
        else
            b = new PNB();

        System.out.printf(
            "RBI rate of interest is : %.0f%%",
            b.rate()
        );
    }
}

```

Compilation and Execution

File Name: Bank.java

```

javac Bank.java
java Bank

```

Sample Execution

```

Enter the Bank name to find the rate of Interest : SBI
RBI rate of interest is : 7%

```

Other Possible Outputs

```

ICICI:
Enter the Bank name to find the rate of Interest : ICICI
RBI rate of interest is : 6%

```

```

PNB:
Enter the Bank name to find the rate of Interest : PNB
RBI rate of interest is : 5%

```

```

RBI:
Enter the Bank name to find the rate of Interest : RBI
RBI rate of interest is : 4%

```

Concept Used: Runtime Polymorphism

The statement `RBI b;` creates a reference of the parent class. It can refer to different child-class objects such as `new SBI()`, `new ICICI()`, and `new PNB()`. When `b.rate()` is executed, Java calls the overridden `rate()` method according to the actual object assigned to `b`. This is an example of runtime polymorphism.

Result

The Java program was successfully implemented to find the rate of interest of different banks using inheritance, method overriding, and runtime polymorphism. For the given input SBI, the displayed rate of interest is 7%.

Conclusion

Thus, the program successfully demonstrates inheritance and method overriding in Java. The SBI, ICICI, and PNB classes inherit from the RBI class and provide their own implementation of the `rate()` method. The appropriate interest rate is displayed based on the bank name entered by the user.